Following the Flow of Molecular Gas in the Galaxy

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This project looks into how the Galactic dynamics, in particular the differential rotation of the disk, might be imprinted in the gas kinematics of molecular clouds. This is investigated using a large survey of the Galactic plane probing the ($J = 2$–1) transition of ^{13}CO. The emission line of the molecular clouds found in the Structure, Excitation and Dynamics of the Inner Galactic Interstellar Medium (SEDIGISM) survey were used to estimate global velocity gradients of the clouds, the direction of which was compared to the direction expected from a cloud dominated by Galactic dynamics, give or take 15°. $\sim 9\%$ of the clouds were found to match this direction, which is indicative of random velocity gradient directions because the percentage matches the 30° range used. Therefore, the clouds at these scales show no significant evidence of imprinted kinematics from the Milky Way.

Introduction

The Interstellar Medium

The interstellar medium (ISM) is made up of primarily gas, as well as dust and cosmic rays. The ISM can be divided up into 5 main phases based on its temperature, ionisation state and density: the cold molecular medium (CMM), where the gas is molecular with a temperature range of $10 - 30\ K$ and a number density above $10^2\ cm^{-3}$; the cold neutral medium (CNM), where the gas is mostly atomic with a temperature range of $30 - 100\ K$ and a number density between $1 - 100\ cm^{-3}$; the warm neutral medium (WNM), where the gas is mostly atomic with a temperature range of $6000 - 10000\ K$ and number densities below $1\ cm^{-3}$; the warm ionised medium (WIM), where the gas is mostly ionised with a temperature range of $8000 - 10000\ K$ and number densities around $0.4\ cm^{-3}$ and, finally, the hot ionised medium (HIM), where the gas is mostly ionised with a temperature range of $10^6 - 10^7\ K$ and number densities below $10^{-2}\ cm^{-3}$ (see Klessen and Glover 2016). The focus of this work will be on the CMM.

Other than primordial elements, every atom formed inside a star. Therefore, initially, after the gas formed in the early universe it was in the HIM, the same is true for the heavier gas formed inside stars. To get to the temperature of the CMM, the gas must be cooled. As the universe expands or as gas moves away from its progenitor, the number of ionising photons will decrease which will offset the balance between ionisation and recombination allowing for the gas to cool to the WIM. At the temperatures of the WIM, the cooling from Lyman-α becomes important. The general form of the cooling rate of a two-level system is shown below.

$$\Lambda_{ul} = \frac{g_u}{g_l} A_{ul} E_{ul} n_l e^{-E_{ul}/k_B T} \tag{1}$$

Equation 1 is the rearranged form of 2.38 in Tielens 2005. The formula comes about by assuming LTE and that the number density is much less than the critical density. g_u and g_l are statistical weights of the energy states, A_{ul} is the Einstein A coefficient of spontaneous emission, E_{ul} is the energy of the transition, and n_l is the number density of the lower state.

$$\Lambda_{HI} = 7.3 \times 10^{-19} e^{-118400/T} n_e n_H \tag{2}$$

Equation 2 appears as 2.70 in Tielens 2005 and shows the cooling rate due to Lyman-α. Where n_e and n_H are the number density of electrons and hydrogen, respectively. The assumption being that collisions between hydrogen and electrons dominate here. Above $10^4 K$, the exponential term stops being negligible, which causes the cooling rate to rise rapidly. This together with recombination to form atomic hydrogen is what facilitates the transition to the WNM. At higher metallicities the cooling is not dominated by Lyman-α but by other fine structure transitions from heavier elements. Once cooled, Equation 2 shows a ceiling for the temperature in the WNM above which the cooling rate can rapidly exceed the different heating rates (Tielens and Hollenbach 1985). A different fine structure transition dominates the cooling in the WNM and facilitates the transition to the CNM, the CII transition. The energy of this transition is $92K$ (Hollenbach and McKee 1989), meaning the exponent in Equation 1 becomes $e^{-92/T}$, allowing the gas to cool efficiently down to $\sim 100K$. The term in the exponent implies that below $100K$ the cooling from CII decreases rapidly. If the density is high enough, molecular gas can begin to form around this temperature. The first rovibrational transition ($J = 1–0$) of CO emits a photon with a rest frequency of $115.27GHz$ (Winnewisser et al. 1997), which gives an energy of $5.53K$, using $h\nu/k_B$. This allows the cooling rate of CO to dominate in this temperature regime and allow the gas to get to a temperature of around $10K$. Even though CO is a more efficient coolant, CII seems to be a more dominant cooling mechanism at low metallicity as well as for diffuse gas with densities less than $10^3 cm^{-3}$ (Glover and Clark 2012). This is partly because at low metallicity, the CO cannot be shielded by the dust and photodissociates by way of the ISRF. It is also important to note that Glover and Clark 2012 indicate that if the cloud is shielded from the effects of photoelectric heating then cooling from CII can allow the gas to get down to temperatures of $20K$ without the presence of CO.

Once the gas in the CNM is sufficiently dense, molecular hydrogen can begin to form. Molecular hydrogen should form directly between two atoms of hydrogen but the Einstein A coefficient is too small for the process to allow for a photon to be emitted during their collision (Wakelam et al. 2017). This means that molecular hydrogen cannot form efficiently this way, therefore another mechanism is required to get the amount of molecular hydrogen expected. The formation process involves atomic hydrogen sticking to a dust grain and colliding with another stuck atom of hydrogen allowing for the recombination of molecular hydrogen. The energy produced by this is enough to break the bonds between the dust grain and the molecule of hydrogen, allowing for it to return to the CMM (Wakelam et al. 2017). This has been confirmed experimentally using various dust grain substitutes made of carbon or silicates. Katz et al. 1999 find that the peak efficiency of the recombination occurs below $20K$ for both types of grain. The formation of CO is very complex (see Glover et al. 2010) and can involve different pathways, the simplest of which is the formation from a carbon atom and a hydroxyl radical. Even though it is thought that the gas being in a molecular state is not necessary for star formation, the cloud being able to shield itself from the ISRF is paramount to the formation of both stars and molecular gas (Glover and Clark 2012). They find that a cloud of atomic gas is still capable of forming stars at the same rate as molecular gas is able to form. This however does not change the fact that molecular hydrogen still traces star formation, it just may not trace all star formation.

Tracing Molecular Clouds

Regions of the ISM that make up the CMM are also known as molecular clouds. These clouds are the densest constituents of the ISM, the collapse of which leads to star formation. Tracing molecular gas is important because it allows for the understanding of the steps that lead to star formation. CO is regarded as a good tracer of molecular gas for its abundance in molecular clouds and spontaneous low energy transitions, like the one described above. The idea is to trace the more abundant molecular hydrogen, which is not directly possible due to its lack of emission at the typical temperatures of the CMM. The lowest possible rotational transition of molecular hydrogen is at an energy of $510K$ (Wakelam et al. 2017). CO has been the gold standard when it comes to tracing molecular hydrogen, but tracers like HCN or N_2H^+ have been found to trace the denser gas much more effectively than CO(Kauffmann et al. 2017). This is because at lower temperatures the CO freezes out and N_2H^+ can only from once CO is no longer present. Additionally, at high densities, CO becomes optically thick because it self-absorbs any photons it emits. N_2H^+ is almost always being detected above column densities of $10^{22} cm^{-2}$ (Fehér et al. 2025), meaning it is much better at tracing the clumps where stars are likely to be forming.

Cloud Collapse

For a cloud to collapse the gravitational force pulling the gas together must overcome that of the internal pressure of the gas, where the internal pressure is assumed to be purely thermal for this derivation. Hydrostatic equilibrium is the point at which both forces cancel one another out. This allows for the derivation of the mass above which a spherical system would collapse, known as the Jeans mass.

$$M_J = \frac{\pi^{3/2} {c_s}^3}{{\rho_o}^{1/2} G^{3/2}} \tag{3}$$

Equation 3 appears as the rearranged form of 9.23 in Stahler and Palla 2004 using the fact that $M_J \sim \rho_o {\lambda_J}^3$, where ρ_o is the initial density before collapse and λ_J is the Jeans length. c_s refers to the isothermal sound speed in the medium given by $\sqrt{k_B T/\mu m_p}$, where k_B is the Boltzmann constant, T is the temperature, μ is the mean molecular weight and m_p is the mass of a proton. This equation assumes that the system is isolated with no external forces, but the ISM is turbulent therefore an external pressure could be present. In this instance the external pressure would oppose the internal pressure of the gas, making it easier for the gas to collapse. This mass is known as the Bonner-Ebert mass.

$$M_{BE} = \frac{1.18 {c_s}^4}{{P_o}^{1/2} G^{3/2}} \tag{4}$$

Equation 4 appears as 9.16 in Stahler and Palla 2004, where P_o is the external pressure on the cloud and G is the gravitational constant. There are several mechanisms that affect star formation that these equations do not account for. Magnetic fields oppose the gravitational collapse of clouds, slowing or stopping the collapse of clouds (Krasnopolsky and Gammie 2005). Turbulence also plays a large role (see below). This allows more gas to be accreted before collapse leading to higher mass star formation. Stellar feedback strips the diffuse gas from the cloud slowing their growth but allowing the dense centres to continue to be star forming (Watkins et al. 2025). It can also cause the destruction of the host cloud. Watkins et al. 2025 also found that stellar feedback likely breaks apart larger clouds into smaller clouds. Protostellar radiative feedback stops fragment formation causing most of the gas to be accreted onto a few objects leading to higher mass star formation (Krumholz, Klein, and McKee 2007). Cloud-cloud collisions between both atomic and molecular clouds facilitates the formation of massive molecular cloud cores, making them an important mechanism for forming high mass stars (Fukui et al. 2021).

As a cloud collapses, the Jeans mass scales proportionally with temperature and density $\left(M_J \propto \rho^{-1/2} T^{3/2}\right)$, as shown by Equation 3. During gravitational collapse the density always increases and providing the gas is isothermal, the Jeans mass decreases as the density increases (Ward-Thompson and Whitworth 2011). Therefore, sections of the collapsing cloud become Jeans unstable and begin gravitational collapse causing multiple collapsing regions. These collapsing regions can go on to fragment again. Eventually the density increases so much so heat is unable to radiate away and cooling becomes unable to keep the gas isothermal, and so temperature increases. Assuming the collapsing cloud acts like a polytrope, it will follow the equation of state $P = K\rho^\gamma$, where γ is the adiabatic index (Chieze 1987). If heat cannot radiate away then the collapse can be considered adiabatic, so the temperature is now a function of density $\left(T = K\rho^{\gamma-1}\right)$. This now means that the Jeans mass only scales proportionally with density $\left(M_J \propto \rho^{(3/2)(\gamma-4/3)}\right)$. In molecular clouds, the gas is primarily diatomic or monatomic which have 5 and 3 d.o.f., which gives them adiabatic indices of 5/3 and 7/5, respectively $(\gamma = 1 + 2/d.o.f.)$. Both these indices result in a direct proportionally between the Jeans mass and density, causing fragmentation to cease.

The above picture of fragmentation is not complete as it does not take into account turbulence or magnetic fields. Magnetic fields as described above stop fragmentation by exerting a force on charged particles outwards. The charged particles then collide with neutral matter causing their velocities to couple (Krasnopolsky and Gammie 2005). It is for this reason that ambipolar diffusion was introduced, which allows the star forming material to be decoupled from the ions affected by strong magnetic fields. This occurs when the clouds get sufficiently dense that cosmic rays and the ISRF are unable to cause substantial ionisation, which reduces the rate at which the collisions occur (Mestel and Spitzer 1956). This allows the charged particles to pass through the neutral matter without imposing its velocity, allowing the neutral matter to go on to fragment and form stars.

Ambipolar diffusion is a slow process and is unable to reproduce the numerical formation time of a core from observations ($\sim 1 Myr$) (Ciolek and Basu 2001). This can be solved with the addition of supersonic turbulence (Mac Low and Klessen 2004). The dominant driving process of this turbulence in star forming galaxies is supernovae, which self regulates by temporarily suppressing star formation allowing more matter to accrete before collapse, leading to high mass star formation and thus more supernovae. These high mass stars during their lifetimes produce ionising radiation that contributes short lasting turbulence by heating the gas which then contracts as it cools (Kritsuk and Norman 2002). Gravitational instabilities are a local driving mechanism of the turbulence but are unable to delay collapse for much more than a free-fall time and so cannot be the dominant driving mechanism. Magnetorotational instabilities allow for turbulent flows to be generated from shear by magnetic fields generating Maxwell stress, which may occur at the outskirts of the Milky Way disc. The supersonic turbulence generated by these driving forces builds up gas at its shock front, forming dense sheets and filaments. These driving forces produce random motions in the gas that can act against gravitational collapse temporarily slowing the infall. This supersonic turbulence can only do so as long as it is continuously driven because it decays quickly. For an isothermal shock the density scales with the Mach number $\left(\rho' = \mathcal{M}^2\rho\right)$, where the Mach number is directly proportional to the turbulent velocity dispersion. The Mach number is how many multiples of the sound speed your gas is travelling at. If the turbulence is treated as an additional pressure then the effective sound speed can be defined which takes into account the turbulent velocity dispersion ($c^2_{\mathrm{s,eff}} = c_s^2 + \sigma^2_{\nu,\mathrm{turb}}/3$) (Chandrasekhar 1951). Velocity dispersions of observed molecular clouds are on the order of a few kms^{-1}. Therefore, it is assumed that $\sigma_{\nu,\mathrm{turb}} >> c_s$ because the turbulence is supersonic, which means that the Jeans mass now is a function of the turbulence as well as the density $\left(M_J \propto \sigma^3_{\nu,\mathrm{turb}}\rho^{-1/2}\right)$. This means that an increase in turbulence increases the Jeans mass, preventing collapse. Eventually the turbulence decays and the cloud can collapse. If the clump is able to become sufficiently dense that it decouples from the ambient gas in the cloud, a second shock will be unable to disrupt it. If the gas is not able to collapse in this time then a second shock would disrupt the clump inducing further turbulence and preventing collapse further. These successive shocks in a short amount of time is the basis for high mass star formation in this theory (see Mac Low and Klessen 2004).

Project Work

It is apparent from the introduction that different physical processes have a direct impact on star formation and affect its rate as well as the mass of the stars forming. Some of these processes can cause changes to the kinematics of the molecular clouds. Before any change in the kinematics due to these processes, the gas is expected to follow the flow imparted by the galaxy during its formation. At what stage during star formation do the dynamics of the galaxy no longer dominate, and the effects of one or more of the processes described above are being seen? The aim of the following work is to answer this question.

The obvious place to conduct this research is in molecular clouds in the Milky Way, due to their proximity, meaning high resolution surveys can be conducted on the molecular clouds. To probe different levels of molecular clouds, different density tracers, which are only present at certain stages of star formation, can be used. ^{12}CO is thought to be present throughout the bulk of molecular clouds but due to its high abundance, it is optically thick in the denser regions making a good tracer for the more diffuse gas at the edges of molecular clouds. ^{13}CO, as an isotopologue, is less abundant, making it better at tracing the denser material in molecular clouds. By being sensitive to the extent of the entire cloud, it is often used to estimate the total column density of molecular gas. Other tracers, such as HCN, trace the intermediate-to-high dense gas in the cloud, while N_2H^+ traces the very high density gas in clumps that are often already actively collapsing.

Data

The SEDIGISM survey (Schuller et al. 2021) traces the (J = 2–1) transition of ^{13}CO, as well as several other tracers not discussed here. This project starts with making use of this survey. The SEDIGISM survey covers typically 0.5° in galactic latitude(b) either side of the galactic plane and 77° in galactic longitude(ℓ) across mainly the fourth quadrant and part of the first quadrant ($300° \leq \ell \leq 18°$). The observations are spectral cubes where the spectral axes is velocity, which ranges from -200 to 200kms^{-1}, with each channel having a width of 0.25kms^{-1}. An individual slice of the spectral cube is typically a size of 2×1 degrees in ℓ and b respectively. To extract the clouds from the survey, Duarte-Cabral et al. 2021 used the Spectral Clustering for Interstellar Emission Segmentation (SCIMES) algorithm (Colombo et al. 2015). The cloud extraction yielded 10663 clouds after dealing with overlapping areas. A final catalogue was produced as well as masks, which contained zeroes everywhere other than the voxels which successfully made it through the cloud-assignment procedure. Those that do not contain zeroes contain the cloud ID.

Moment Maps

For every spatial coordinate, the masks give the vowels that contain significant ^{13}CO(2–1) emission. Therefore, information about the spectrum can be obtained from the voxels in the survey that the masks correspond to. In spectroscopy, different moments can be computed which show different aspects of a spectrum. To that end, code was written to calculate, in an automated way, the zeroth, first and second moment for each individual cloud in the survey, creating output FITS files containing the data for each moment.

$$M_0 = W = \Delta\nu \sum_{i=0}^{n} I_i \tag{5}$$

Equation 5 shows the equation for the zeroth moment, which corresponds to the total integrated flux. n is the number of channels in the cube, which for SEDIGISM is 1600, $\Delta\nu$ represents the channel width in kms^{-1} and I_i represents the flux in an individual voxel, i. Thus, the flux is summed along each line of sight, giving the total integrated intensity. Equation 5 should be interpreted as the area under the emission line, for each spatial coordinate.

$$M_1 = <\nu> = \frac{\Delta\nu \sum_{i=0}^{n} I_i \nu_i}{M_0} \tag{6}$$

Equation 6 shows the equation for the first moment or the intensity weighted velocity of the spectral line, where ν_i is the velocity of channel corresponding to the flux I_i. The zeroth moment is required to calculate the first moment. Equation 6 gives the mean velocity of the ^{13}CO gas in that line of sight.

$$M_2 = \sigma_\nu = \sqrt{\frac{\Delta\nu \sum_{i=0}^{n} I_i (\nu_i - M_1)^2}{M_0}} \tag{7}$$

Equation 7 shows the equation for the second moment or the velocity dispersion of the spectral line. It is dependent on the previous two moments, and it gives the width of the spectral line.

With the moment maps generated, and in particular the moment 1 maps (i.e. the velocity maps), the inner kinematics of clouds and their global kinematic patterns, can begin to be explored. Do clouds from different environments have different preferred direction of spin compared to that of the Galaxy as a whole? This is something that will help to understand if the dynamics of the galaxy are still dominating at these scales and by looking at different environments may indicate that the galaxy dynamics might play a more important role in setting the dynamical properties of molecular clouds. If the spin of the gas in a cloud was following that of the Milky Way, the spin vector would directly oppose that of the direction of rotation of the galaxy and face towards positive latitude. This is due to the gas in the galaxy rotating differentially, causing shear, meaning the clouds would have a diametrically opposing spin direction (see Figure 1). Therefore, by analysing the velocity of these clouds and calculating their large-scale gradient allows for the direction of the spin vector to be derived.

Velocity Gradients

Since the moment 1 map contains the mean velocity for each spatial pixel of the cloud, they will be used to calculate the velocity gradient across the cloud. It is important to note that these velocities in these moment maps are one-dimensional and are not representative of the cloud as a whole. Additionally, there is no information on the inclination of these clouds, so the velocities have not been corrected for this, although this does not directly affect this work. The clouds are all at different distances therefore the physical width of each pixel changes drastically over the clouds in the catalogue. This change in physical scale means that any direct comparison between the kinematics of two clouds at their native resolution would be unreliable. Therefore, the analysis is uniformised by bringing all clouds to a constant physical scale. To do this a Gaussian filter was applied to all moment 1 maps. This Gaussian filter is a two-dimensional convolution that takes input the standard deviation of the Gaussian used to do the convolution. The choice of a kernel to uniformise the sample required careful testing: choosing a kernel much larger than the typical size of clouds, would mean that the gradient could not be computed for the smallest clouds; while choosing a kernel too small would exclude clouds at large distances, for which the physical resolution is much lower than nearby clouds. To decide on which physical scales to consider, the distribution of medial axis sizes from Duarte-Cabral et al. 2021 (their Figure 8), as well as considering the lowest physical resolution probed for the furthest away clouds (e.g. a cloud at $\sim$ 24 kpc has an original resolution of 3.3 pc). 5 pc was taken as our final physical FWHM beam size after the convolution.

$$\Omega_f{}^2 = \Omega_i{}^2 + \Omega_k{}^2 \tag{8}$$

As the original image's beam is described by a Gaussian and the convolution is being completed using a Gaussian function then the resulting image will also be a Gaussian. The variance of the Gaussian can be described by the addition in quadrature of the contributing Gaussian's variances. As the conversion from variance to FWHM is the same for all three factors in the equation, the equation can be written as shown in Equation 8. Where Ω_i represents the FWHM beam size, which for the SEDIGISM observations is 28 arcseconds (Duarte-Cabral et al. 2021). Working backwards from a final image, the required angular FWHM of the final image (Ω_f) can be calculated, and therefore, the FWHM of the kernel for the convolution (Ω_k) can be inferred. Using the fact that $\sigma = \mathrm{FWHM}/2\sqrt{2\ln 2}$, the standard deviation of the Gaussian kernel is calculated. As the original physical beam sizes of all the clouds was less than 5 pc, there is the added benefit that the data has been smoothed and so the moment 1 maps now show much better how the velocity changes globally.

The velocity gradient is calculated on a pixel by pixel basis and for an individual pixel it is dependent on the velocities of the eight pixels surrounding it. The gradient in galactic longitude and galactic latitude is calculated individually in terms of $\Delta\nu_x$ and $\Delta\nu_y$ and is also converted into a magnitude $\sqrt{\Delta\nu_x^2 + \Delta\nu_y^2}$ and an angle $\tan^{-1}(\Delta\nu_y/\Delta\nu_x) \times (360/2\pi)$, each of these are saved as a FITS image. The velocity gradient is defined as the change in velocity per unit distance and these changes are calculated horizontally, vertically and diagonally, by subtracting opposite velocities from one another. Initially the units are $\mathrm{kms^{-1}pix^{-1}}$, but they are converted to be with respect to the angular size of the pixel in radians $\mathrm{kms^{-1}rad^{-1}}$. The gradient in the direction of galactic longitude is the sum of the horizontal change in velocity and the horizontal components of the diagonal change in velocity. Typically, all surrounding pixels must be non-zero for the gradient to be calculated, but this is often not satisfied for smaller clouds. To mitigate this and maximise the number of points within a cloud where the gradient can be estimated, if any of the diagonal points are missing, the gradient is still estimated based purely on the vertical and horizontal components. In these cases, the absolute gradient vector is not as accurate, but it's still a good approximation of the general direction, which is the main goal. The gradient in galactic longitude and latitude is positive for a gradient in the direction of positive galactic longitude and latitude. The angle is in degrees and is zero for positive latitude and increases clockwise towards negative galactic longitude. The methods for calculating the velocity gradient were used on the raw moment 1 maps as well as the many differently smoothed versions and after comparison of several clouds it was obvious that the smoothed moment 1 maps give a much more precise velocity gradient.

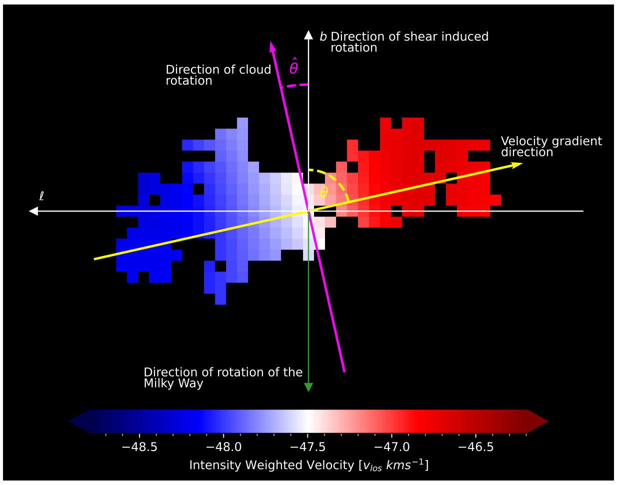

Figure 1: Moment 1 map of a SEDIGISM cloud showing the direction of velocity gradient, and how it compares to the direction of the shear-induced rotation, from the differential rotation of the Galaxy. The moment 1 map is cloud 4107 from the SEDIGISM catalogue and has been is smoothed to a physical resolution of 5pc.

The angle of the velocity gradient of the cloud as a whole, indicated by the yellow arrow in Figure 1, was calculated by taking the mean of all the pixel by pixel velocity gradients in galactic longitude and latitude and converting to an angle and magnitude which are stored in an updated version of the catalogue. Due to the intricacies of errors when it comes to a cyclical values like angles, the errors could not be well constrained and so an estimate of the spread of the angles has been made. This was done by taking the circular standard deviation of the angles in the FITS image showing the direction of the velocity gradients pixel by pixel. The velocity gradient could not be estimated for all clouds as some of them had too few pixels with values in the moment 1 map to have a sufficient number of surrounding pixels for the calculation to take place. The velocity gradients for 10293 clouds were successfully calculated.

Using only the immediate pixels surrounding a point for the calculation of the velocity gradient is technically incorrect because the image has been smoothed. This is due to the nature of convolutions, and it means that a given pixel, post smoothing, is now dependent on the pixels half the kernel length away in radius. The kernel's length is a function of the Gaussian's standard deviation and scales as $l_k = 2 \times \mathrm{int}(4\sigma) + 1$. This means that the kernel can be rather large, but as the Gaussian produced for the convolution extends as far as 4σ the weights at the edges of the kernel will be negligible, therefore these outer pixels will not have much of an effect on the smoothed value. If the significant bandwidth of this Gaussian is assumed to be σ, then based on the Nyquist sampling theorem, the diameter for the surrounding pixels should be 2σ, to avoid aliasing. The issue with using this method is that the velocity gradient cannot be calculated for over half of the clouds due to not having pixels with values in at these distances. Therefore, both methods were tested to compare the direction of the velocity gradients for the half that could be calculated.

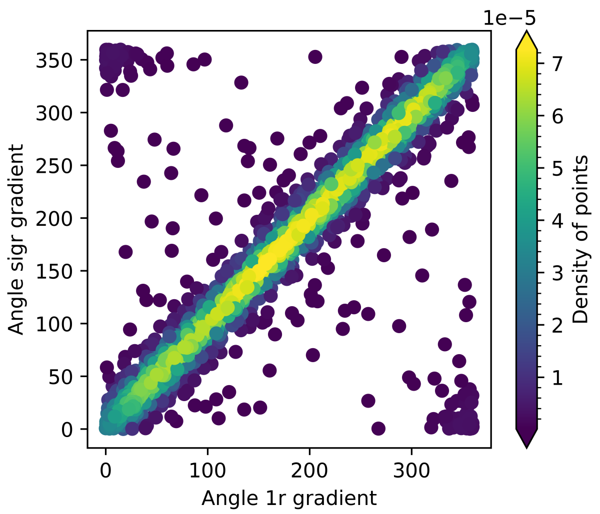

Figure 2: The comparison of the results from using the different methods for calculating the direction of velocity gradient.

Figure 2 shows a positive linear correlation with a Pearson's correlation coefficient of 0.86. The density of points has been included to show that the trend is more linear that it may appear. The group of points at $(0, 360)$ and $(360, 0)$ realistically should fall on the straight line, but due to the cyclical nature of angles, the mean can fall just above zero and still be very close to 360°. These edge cases were not removed during the calculation of Pearson's correlation coefficient, it would be closer to one if the points had been removed. There are a few clouds that have vastly changing directions for the velocity gradient, but a significant majority have matching angles. Therefore, to ensure that almost all the clouds have their velocity gradients calculated, the method will go forward using the radius of one pixel for the gradient calculation.

Preliminary Results

For a theoretical cloud dominated by the dynamics of the galaxy, the velocity gradient direction of the gas inside the cloud would be in the galactic plane pointing towards negative galactic longitude. Therefore, if the velocity gradient direction calculated above is approximately 90°, then it would be facing the direction of negative galactic longitude. A range between 75° and 105° was used to classify clouds as having a velocity gradient sufficiently in the plane of the galaxy. It is possible that a cloud that has a velocity gradient in this direction could have its spin induced by galactic shear.

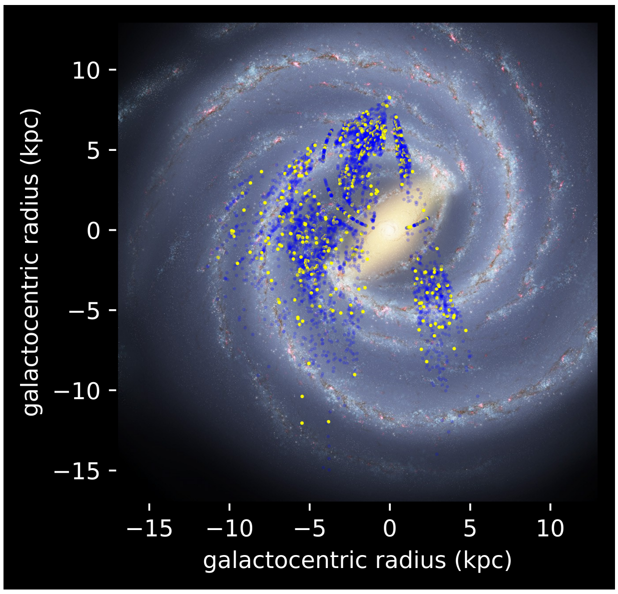

Figure 3: Top-down view of SEDIGISM clouds with reliable distances, coloured by whether they have a velocity gradient consistent with shear-induced rotation (yellow), or not (blue). The Sol system is located approximately 8kpc up from the galactic centre on this image. The picture is an artist's impression of the Milky Way [by NASA/JPL-Caltech/R. Hurt (SSC/Caltech)].

Not all clouds are included in Figure 3 because they do not pass a quality assurance step included before plotting, where the quality assurance ensures that only data that is scientifically accurate is used. Some of the clouds in the SEDIGISM catalogue do not have distances that can be considered reliable, meaning the methods used are not reliable. Most of the clouds that do not have reliable distances associated with them are located in the centre of the galaxy, which is why in Figure 3 no points appear in the galactic centre. Whether the distance is reliable is already in the catalogue under the term d_reliable and is equal to one when reliable. The other quality assurance test checks if the direction of the velocity gradient is a precise measurement, this is done by ensuring the spread, as discussed above, is less than 40°. This number has been chosen arbitrarily and serves as a way of ensuring the velocity gradients are pointing in all directions, which one would expect from a cloud within which the gas is not rotating.

The fraction of clouds thought to be shear induced to the total number of clouds is found to be $\sim 9\%$. This is indicative of random cloud spin directions because the proportion of the range used to define clouds as shear induced compared to the total range is $30°/360° \approx 8.3\%$. Since the velocity gradient direction is random, the $\sim 9\%$ of clouds that were considered to have their rotation induced by shear can no longer be. The mechanism which is driving the velocity gradient across the clouds is not known, but this result shows that, at the scales shown by ^{13}CO, the velocity gradient is not dominated by the dynamics of the galaxy.

Future Work

The next step in this project is to look at a more diffuse tracer of molecular clouds and repeat this test to see if the molecular gas is dominated by the dynamics of the galaxy at larger scales. ^{12}CO as stated at the beginning of this section is one such tracer and a survey exists already which contains such data, the Three-mm Ultimate Mopra Milky Way Survey (ThrUMMS) (Barnes et al. 2015). Unfortunately, the cloud extraction for this has not yet been completed therefore, to use the data, the next objective is to use SCIMES (Colombo et al. 2015) to complete the cloud extraction. Additionally, the methods used to calculate the velocity gradient must be compared to methods which exist in the field currently (Arquilla and Goldsmith 1986), to ensure the precision of this work.

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